Math 152 Mon 4/1/24 Exam II

Wich of the following sequences is not convergent

$$\lim_{n\to\infty}\left(\frac{5^n+2}{3^n}\right) = \frac{5^n \ln 5}{3^n \ln 3}$$

$$\frac{5^n (\ln 5)^2}{3^n (\ln 3)^2}$$

a) $a_n = \dfrac{5^n+2}{3^n}$ ← correct answer

b) $a_n = \dfrac{4n-1}{2n+1}$ ← why is this convergent

c) $a_n = \dfrac{3}{n!}$

d) $a_n = \ln(n+2) - \ln(n+3)$ $\quad \lim_{n\to\infty} \ln\left(\dfrac{n+2}{n+3}\right) = 0$

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test for convergence (abs or cond) or divergent

$$b_n = \frac{1}{\sqrt{n^2+3n+1}} \qquad \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+3n+1}} \leftarrow \text{last}$$

$$\lim_{n\to\infty} \frac{1}{\sqrt{n^2+3n+1}} = 0$$

limit comparison Test

① LCT ② Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ which a divergent p-series $p=1 \geq 1$

③ $$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2+3n+1}} \cdot \frac{\sqrt{n^2}}{1}\right) = \lim_{n\to\infty}\left(\sqrt{\frac{n^2}{n^2+3n+1}}\right) = \sqrt{1} = 1$$

$1 > 0$ and finite

then they are BOTH Divergent

$$\boxed{\text{AST: } \lim_{n\to\infty} b_n = 0 \checkmark \quad \text{ii } b_{n+1} < b_n \checkmark}$$

so $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+3n+1}}$ is Conditionally convergent

* What's a good way/easy way to memorize these formulas

Practice

* When should you use an <u>Alternating Series Test</u>

after you use another test first

$a_n = (3n+1)!$ $\quad a_{n+1} = (3(n+1)+1)!$

$(3n+4)!$

$(3n+1)!\,(3n+2)(3n+3)(3n+4)$

Difference between absolute convergence and conditional convergent; how to identify.

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$ $b_n = \frac{1}{n^3}$ which is a convergent p-series so I don't need the Alt. Series Test — absolutely convergent

Ratio test

$\sum_{n=1}^{\infty} \frac{(5 \cdot 10 \cdot 15 \cdots (5n))}{n!}$ $(5(n+1))$

$a_{n+1} = \frac{(5 \cdot 10 \cdot 15 \cdots (5n)(5n+5)}{n!(n+1)}$

$\lim_{n \to \infty} \left| \frac{\cancel{(5 \cdot 10 \cdot 15 \cdots (5n)}(5n+5)}{(n+1)\cancel{n!}} \cdot \frac{\cancel{n!}}{\cancel{(5 \cdot 10 \cdot 15 \cdots (5n)}} \right|$

$\lim_{n \to \infty} \left| \frac{5n+5}{n+1} \right| = 5 > 1$ so Divergent

$2^{n+1} = 2^n \cdot 2$

Ratio test

$\sum_{n=1}^{\infty} \frac{(-2)^n}{(2n+1)!}$ $\frac{(-1)^n 2^n}{(2n+1)!}$ ignore

$a_{n+1} = \frac{2^n \cdot 2}{(2n+1)!(2n+2)(2n+3)}$

$(2(n+1)+1)! = (2n+3)!$

$\lim_{n \to \infty} \left| \frac{\cancel{2^n} \cdot 2}{\cancel{(2n+1)!}(2n+2)(2n+3)} \cdot \frac{\cancel{(2n+1)!}}{\cancel{2^n}} \right|$

$\lim_{n \to \infty} \left| \frac{2}{(2n+2)(2n+3)} \right| = 0 < 1$ $\therefore$ absolutely convergent

B. Find the length of the Curve $x = \ln(y^2-1)$ $2 \le y \le 3$

$$L = \int_c^d \sqrt{1+(dx/dy)^2}\,dy$$

$$\frac{dx}{dy} = \frac{1}{y^2-1}(2y) \qquad \left(\frac{dx}{dy}\right)^2 = \frac{4y^2}{(y^2-1)^2}$$

$$L = \int_2^3 \sqrt{1+\frac{4y^2}{y^4-2y^2+1}}\,dy = \int_2^3 \sqrt{\frac{y^4-2y^2+1+4y^2}{y^4-2y^2+1}}\,dy \qquad (y^2+1)^2$$

$$L = \int_2^3 \sqrt{\frac{(y^2+1)^2}{(y^2-1)^2}}\,dy = \int_2^3 \frac{y^2+1}{y^2-1}\,dy$$

long division first!

$$\begin{array}{r} 1 \\ y^2-1 \overline{\big) y^2+1} \\ -(y^2-1) \\ \hline 2 \end{array}$$

$$L = \int_2^3 1 + \left(\frac{2}{y^2-1}\right) dy$$

partial fraction

look at other paper for B

2. Are the following improper Integrals Convergent or divergent? Describe What Convergence means.

$$\int_0^3 \frac{4x}{\sqrt{9-x^2}}\,dx$$

only problems are $x = \pm 3$

$u = 9-x^2 \qquad du = -2x\,dx \qquad \frac{du}{-2x} = dx$

$$\lim_{t\to 3} \int_0^t \frac{4x}{\sqrt{9-x^2}}\,dx \qquad \lim_{t\to 3} \int_0^t 4x\,(u)^{-1/2}\left(\frac{du}{-2x}\right)$$

$$\lim_{t\to 3} -2\int_0^t -2u^{-1/2}\,du = \lim_{t\to 3} -2\left(\frac{1}{\frac{1}{2}}u^{1/2}\right)\Big|_0^t = \lim_{t\to 3} -4\sqrt{9-x^2}\Big]_0^t$$

$$\lim_{t\to 3}\left[-4\sqrt{9-t^2} - (-4\sqrt{9}\right] =$$

$$-4\sqrt{0} + 4(3) = 12 \text{ so convergent}$$

3. Use Integral Test to determine if the Series is divergent Or convergent. Find the remainder estimate for S_{10}

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}\,4^{\sqrt{n}}}$$

Alternating Series Estimation Theorem

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n+3}$$

Remainder after $|S_{10}|$

$$|R_{10}| \le b_{n+1} \le b_{11}$$

$$b = \frac{1}{n+3}$$

$$|R_{10}| \le \frac{1}{14}$$

$$\frac{1}{10} = \cancel{}$$

$$n+1 = 7$$

$$n = 6$$

$$S_6$$

(B) $\frac{A}{y+1} + \frac{B}{y-1} = \frac{2}{(y+1)(y-1)}$ $\quad A(y-1) + B(y+1) = 2$

$y = 1$ $\quad B = 1$

$y = -1$ $\quad A = -1$

$$h = \int_2^3 \left(1 + \frac{-1}{y+1} + \frac{1}{y-1}\right) dy = y - \ln|y+1| + \ln|y-1| \Big]_2^3$$

$$y + \ln\left|\frac{y-1}{y+1}\right| \Big]_2^3 = \left(3 + \ln\left|\frac{2}{4}\right|\right) - \left(2 - \ln\left(\frac{1}{3}\right)\right)$$